AC communication module

Overview

This document describes the specification of the PZEM-004T AC communication module,

the module is mainly used for measuring AC voltage, current, active power, frequency, power

factor and active energy, the module is without display function, the data is read through the TTL

interface.

PZEM-004T-10A: Measuring Range 10A (Built-in Shunt)

PZEM-004T-100A: Measuring Range 100A (external transformer)

1.Function description

1.1 Voltage

1.1.1 Measuring range:80～260V

1.1.2 Resolution: 0.1V

1.1.3 Measurement accuracy: 0.5%

1.2 Current

1.2.1 Measuring range: 0～10A(PZEM-004T-10A); 0～100A(PZEM-004T-100A)

1.2.2 Starting measure current: 0.01A(PZEM-004T-10A); 0.02A(PZEM-004T-100A)

1.2.3 Resolution: 0.001A

1.2.4 Measurement accuracy: 0.5%

1.3 Active power

1.3.1 Measuring range: 0～2.3kW(PZEM-004T-10A); 0～23kW(PZEM-004T-100A)

1.3.2 Starting measure power: 0.4W

1.3.3 Resolution: 0.1W

1.3.4 Display format:

□1000W, it display one decimal, such as: 999.9W

≥1000W, it display only integer, such as: 1000W

1.3.5 Measurement accuracy: 0.5%

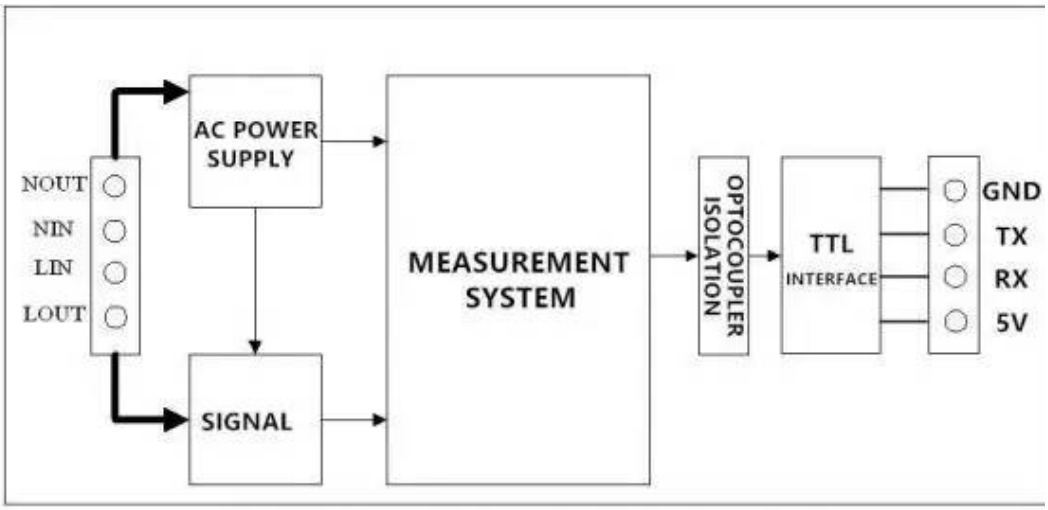
1.4 Power factor

1.4.1 Measuring range: 0.00□1.00

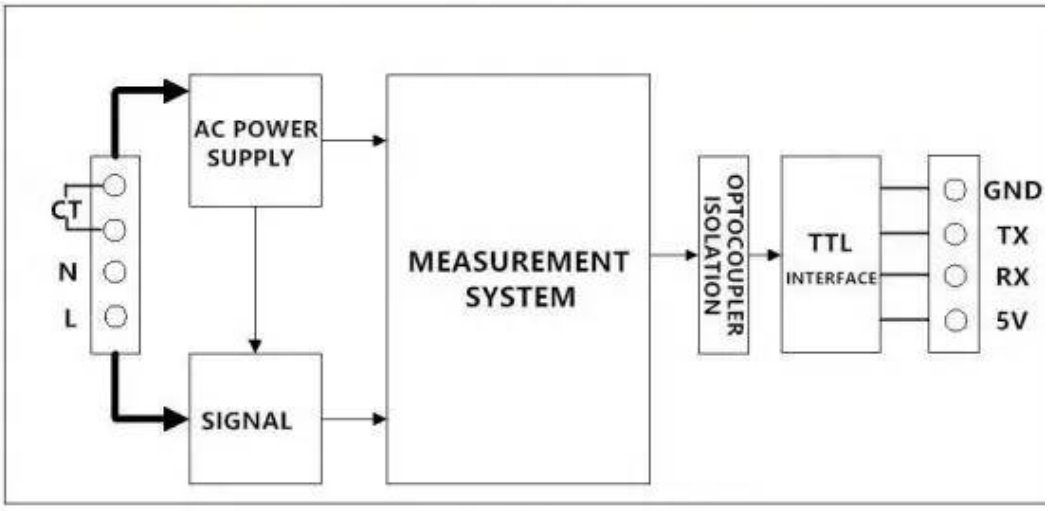
1.4.2 Resolution: 0.01

Register address	Description	Resolution
0x0000	Voltage value	1LSB correspond to 0.1V
0x0001	Current value low 16 bits	1LSB correspond to 0.001A
0x0002	Current value high 16 bits	
0x0003	Power value low 16 bits	1LSB correspond to 0.1W
0x0004	Power value high 16 bits	
0x0005	Energy value low 16 bits	1LSB correspond to 1Wh
0x0006	Energy value high 16 bits	
0x0007	Frequency value	1LSB correspond to 0.1Hz
0x0008	Power factor value	1LSB correspond to 0.01
0x0009	Alarm status	0xFFFF is alarm, 0x0000is not alarm

Register address	Description	Resolution
0x0001	Power alarm threshold	1LSB correspond to 1W
0x0002	Modbus-RTU address	The range is 0x0001~0x00F7



Picture 3.1 PZEM-004T-10A Functional block diagram



Picture 3.2 PZEM-004T-100A Functional block diagram

4 Wiring diagram

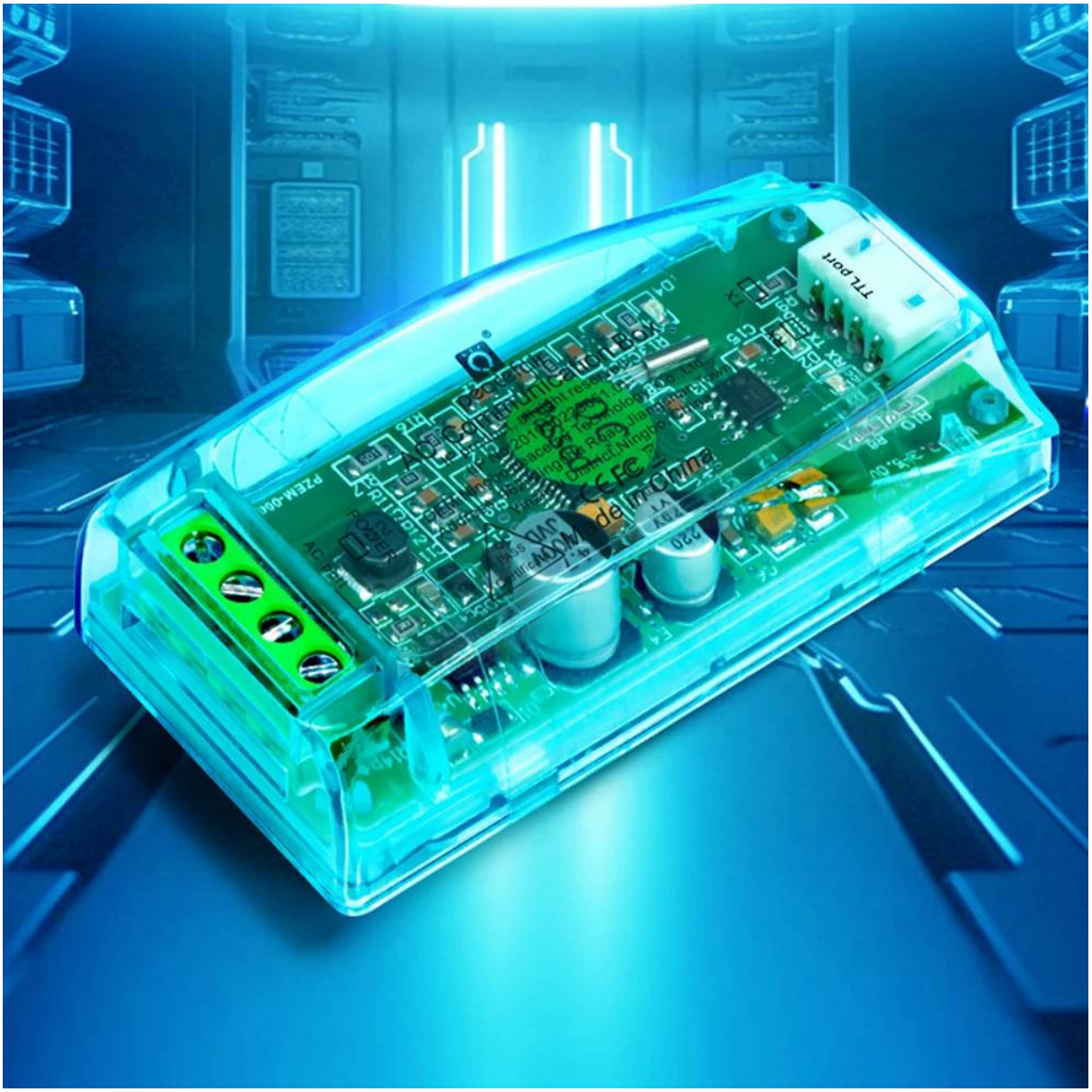


Picture 4.1 PZEM-004T-10A wiring diagram

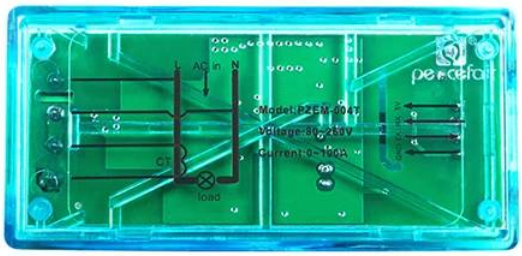


Picture 4.2 PZEM-004T-100A wiring diagram

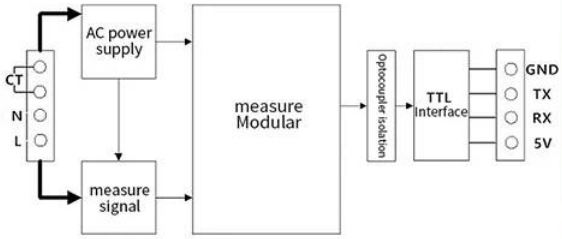
Pictures


TTL port

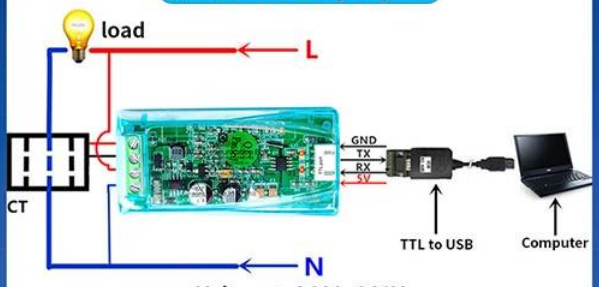
100A range (External Transformer)



PZEM-004T functional block diagram



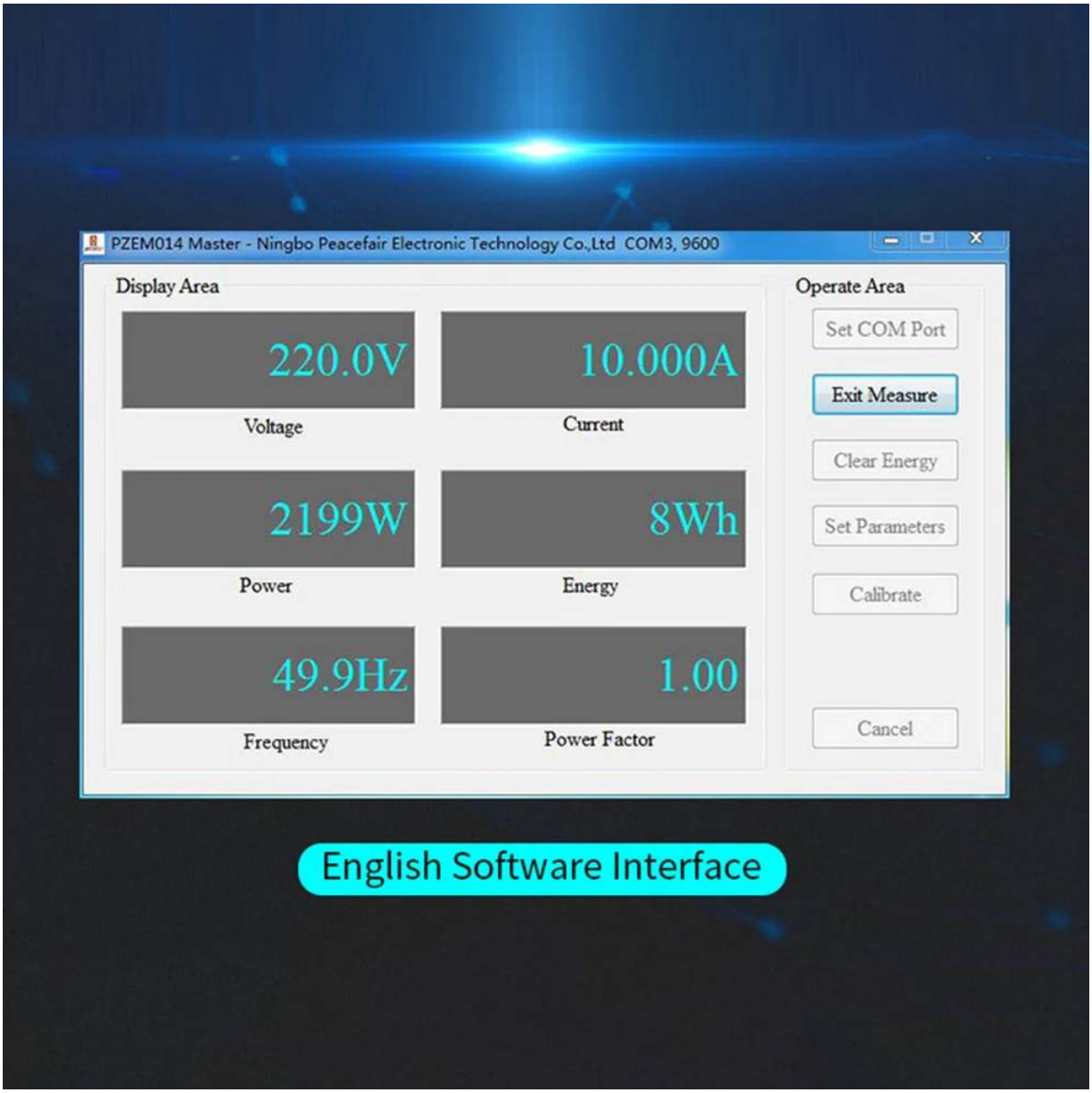
PZEM-004T Wiring diagram



Voltage：AC80~260V
Electric current：0~100A

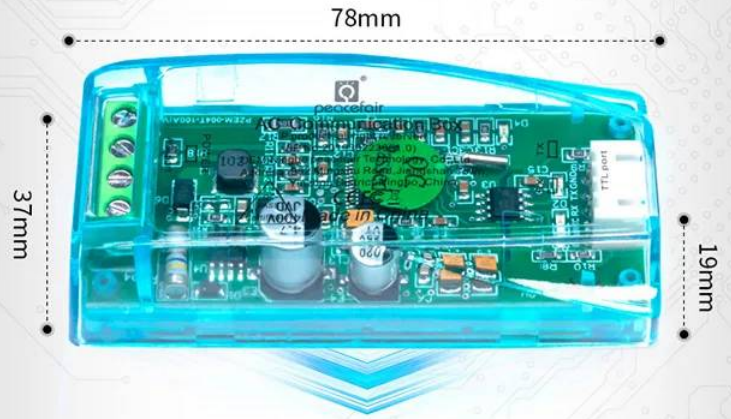

100A range (external transformer)
004T-100A host
Closed CT 100A
LOADING.....
100A range (external transformer)
004T-100A host
Open-close CT 100A
TTL to USB
LOADING.....
100A range (external transformer)
004T-100A host
Closed CT 100A
TTL to USB
LOADING.....


Chinese-English Software Double Interface
Bring you different experiences
PZEM014上位机 - 宁波培正电子科技有限公司 COM3, 9600
显示区域
220.0V
电压
10.000A
电流
2199W
功率
31Wh
电量
50.0Hz
频率
1.00
功率因数
操作区域
通信设置
退出测量
清零电量
参数设置
校准
退出
Chinese Software Interface

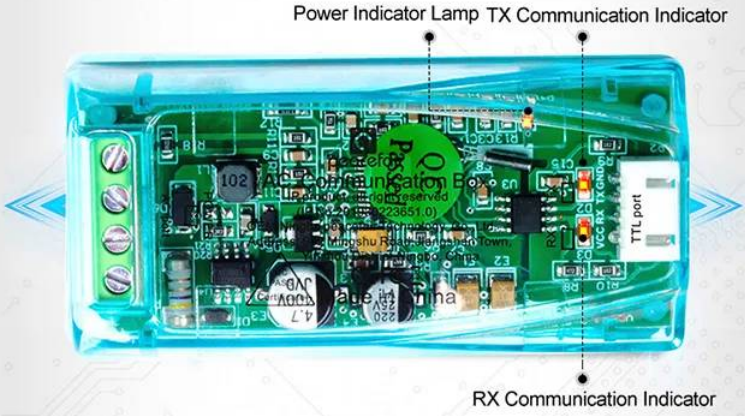

PZEM014 Master - Ningbo Peacefair Electronic Technology Co.,Ltd COM3, 9600
Display Area
220.0V
Voltage
10.000A
Current
2199W
Power
8Wh
Energy
49.9Hz
Frequency
1.00
Power Factor
Operate Area
Set COM Port
Exit Measure
Clear Energy
Set Parameters
Calibrate
Cancel
English Software Interface

Product parameters

Product size



Indicator display



FUNCTION DESCRIPTION

Function	Measuring range	Starting measure current/power	Resolution	Measure-ment accuracy	Display format
Voltage	80~260V		0.1V	0.5%	
Current	0~100A	0.02A	0.001A	0.5%	
Active power	0~23kW	0.4W	0.1W	0.5%	<1000W, it display one decimal, such as: 999.9W; ⩾1000W, it display only integer, such as: 1000W
Power factor	0.00~1.00		0.01	1%	
Frequency	45Hz~65Hz		0.1Hz	0.5%	
Active energy (Reset energy: use software to reset)	0~250000 kWh		1Wh	0.5%	<110kWh, the display unit is Wh(1kWh=1000Wh), such as: 9999Wh; ⩾10kWh, the display unit is kWh, such as: 9999.99kWh
Over power alarm	Active power threshold can be set, when the measured active power exceeds the threshold, it can alarm				
Communication interface	RS485 interface				
size	Length * width * height=73.7*30*14.3mm (Bare pager)				
Power Supply	The power supply of single-phase power-frequency network supplies power to the main circuit through resistance-capacitance step-down, TTL output communication interface and Main circuit optocoupler isolation, for passive output, External power is required for communication				
working temperature	-20°C~+60°C				

SPECIAL NOTE

The TTL interface of this module is a passive interface, which needs external VCC(voltage range 3.3~5V) power supply, that is, the four ports must be connected (VCC, RX, TX, GND) during communication, otherwise the communication cannot be performed.

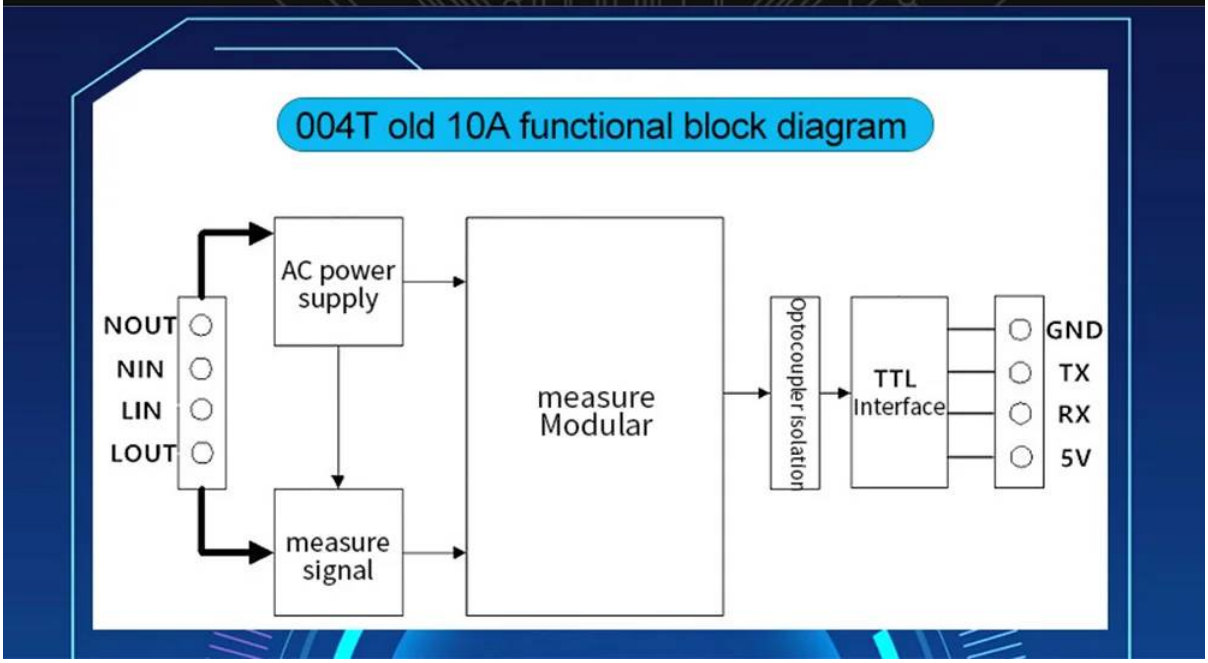
Operating temperature :-20°C~+60°C.

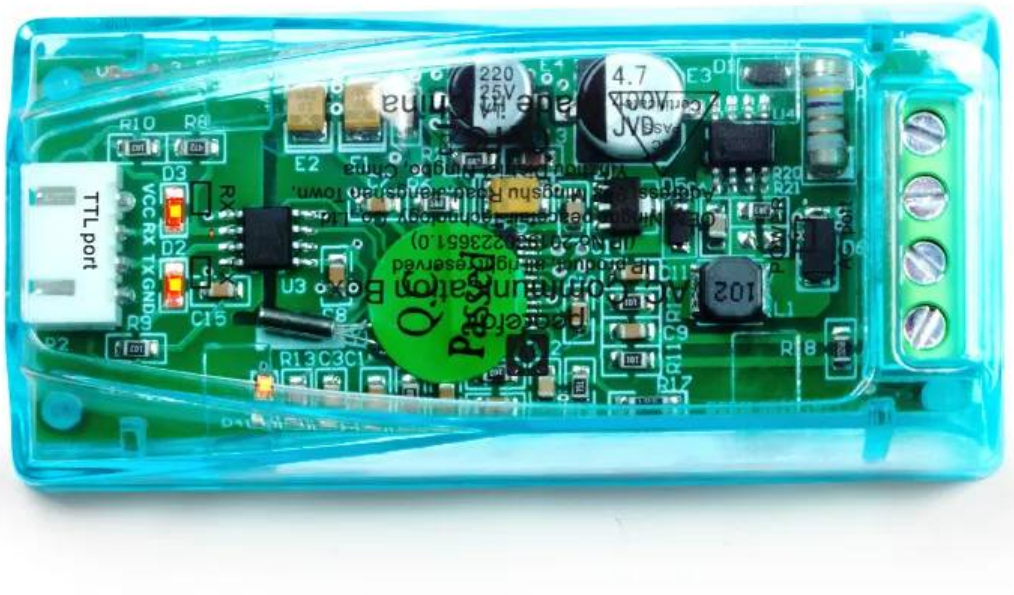

COMMUNICATION PROTOCOL SHOW

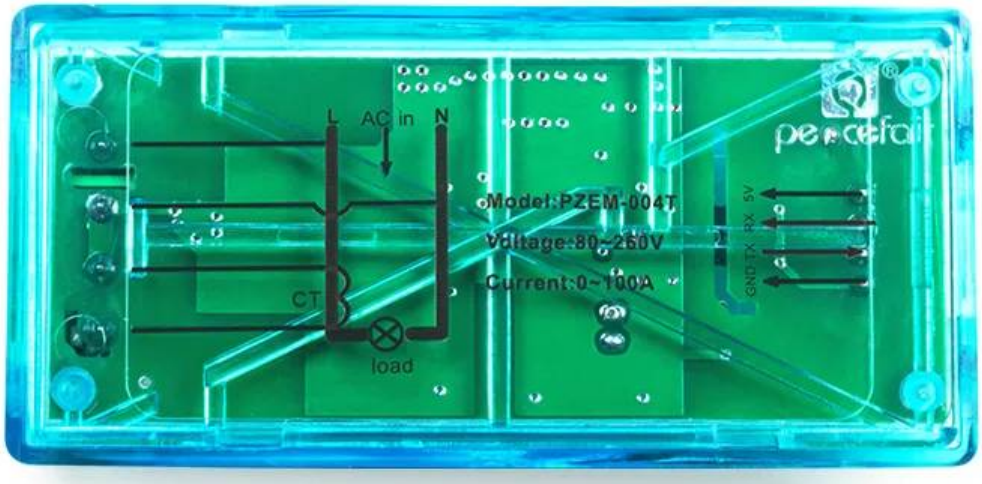
Our company's patented products

Any company, manufacturer orindividual counterfeit products
We will pursue the relevant legal liability according to law
and bear the resulting market losses!

Appearance patent number:
201930223651.0




TTL port
VCC RX TX GND


L
AC in
N
Model:PZEM-004T
Voltage:80~260V
Current:0~100A
CT
load
GND TX RX 5V


Q.C.
Passed
VCC RX TX GND
VCC=3.3-5.0V
PZEM-004T-100A(V4.0)


TTL port



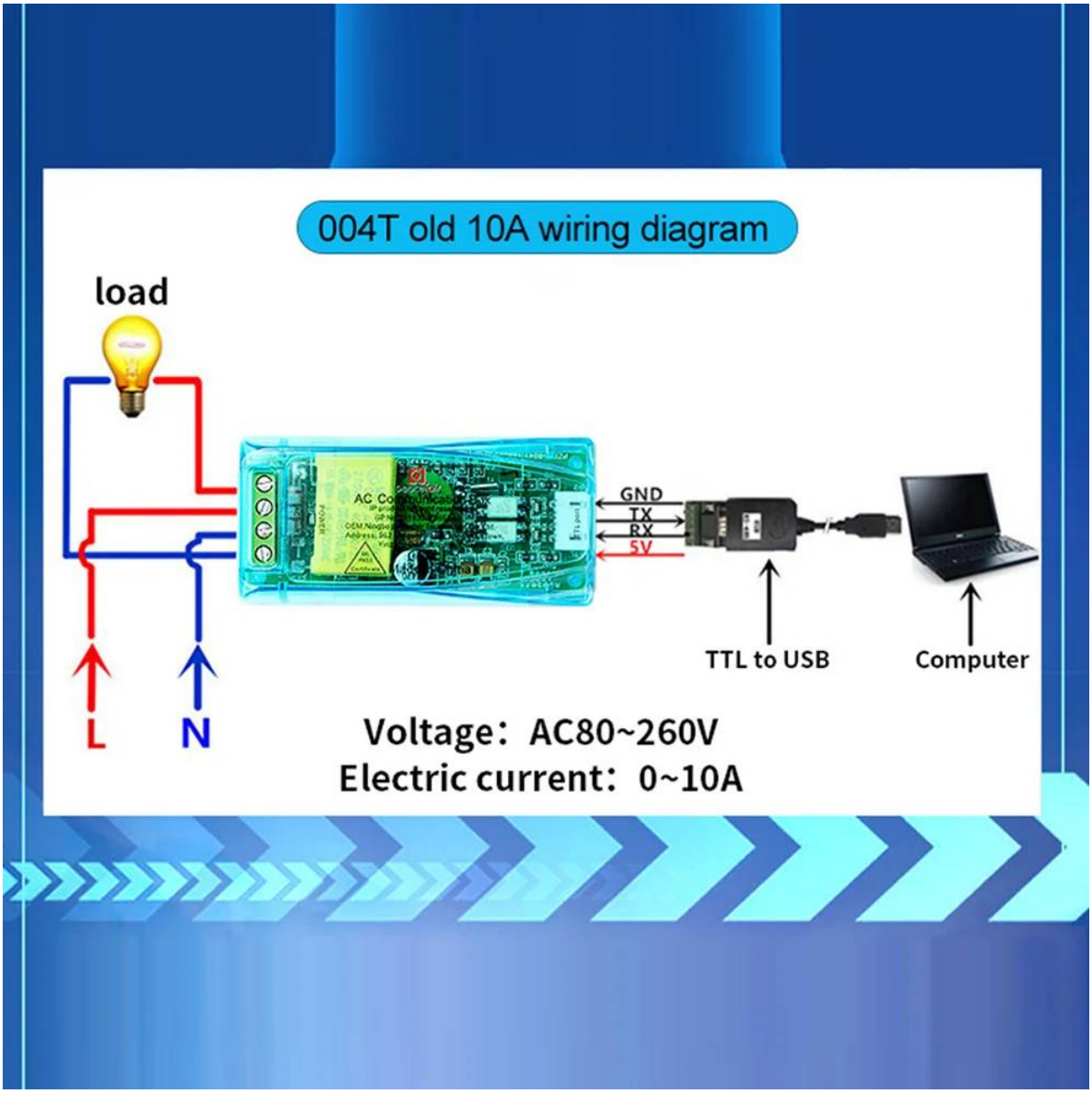

4.7
400V
JVD
220
25V
VT
Q.C.
Passed
PZEM-004T-100A(V4.0)
VCC=3.3-5.0V
P2


TTL port
Communication BOX
Made in China


VCC=3.3-5.0V
220
25V
VT
4.7
400V
JVD
VCC RX TX GND
P2
Q.C.
Passed
PZEM-004T-100A(V4.0)



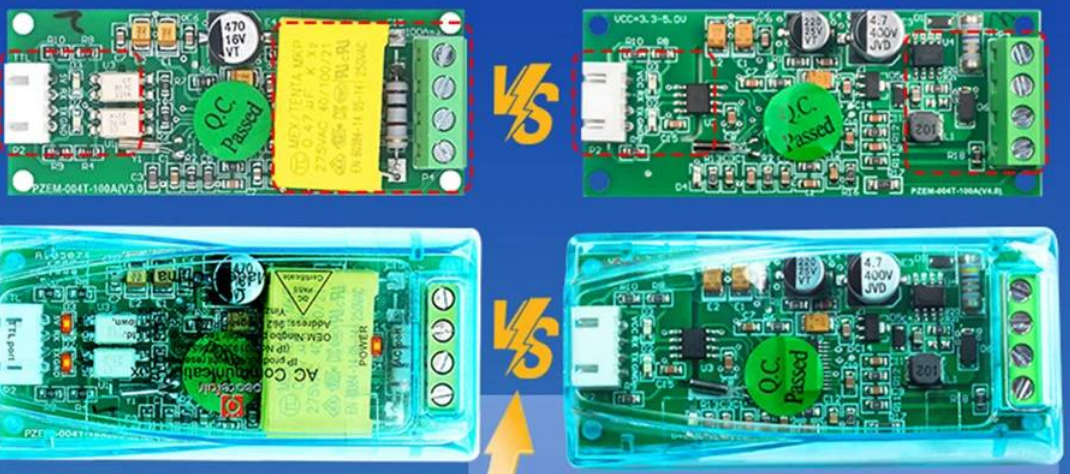

VCC=3.3-5.0V
P2
R10
R8
D3
D2
C15
D4
R13
C3
C2
C1
R1
C14
D5
U4
R20
R21
L1
R18
4.7
400V
JVD
Passed
O.K.
PZEM-004T-100A(V4.0)


004T old 10A wiring diagram
load
GND
TX
RX
5V
TTL to USB
Computer
L
N
Voltage：AC80~260V
Electric current：0~10A


Upgrades 80%
Ac communication module

New upgrade shock release

Higher accuracy, faster response and higher communication stability



004T old version	004T upgrade version
Resistance-capacitance pressure reduction scheme Poor stability, easy to be disturbed	AC-DC digital power supply solution Better anti-interference performance
Single level communication scheme Only 5V serial port communication is supported	Wide level communication scheme Compatible with 3.3 to 5V serial communication
Optocoupler isolation scheme Prone to fluctuations and speed limits	Digital isolation chip solution More stable communication


High precision
fast response
All WIN systems are equipped with upper computer software
TTL port


Fully transparent
blue film housing
TTL port

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SHIPMENT

Shipping Method	Courier	Shipping Time
Express Shipping	DHL Express	About 3-10 days
	EMS Shipping	About 3-10 days
Standard Shipping (for order over USD5)	Aliexpress Standard Shipping	About 20-45 days
	Registered Air Mail(china post/e packet-only for USA)	About 40-60 days
Economy Shipping (for order under USD5)	Economy Post(finland economy/ SG economy/China economy)	About 40-60 days

FEEDBACK



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Item as described : ★★★★★ 5 STAR

Communication : ★★★★★ 5 STAR

Shipping speed : ★★★★★ 5 STAR

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